DATA ITEM DESCRIPTION

Title: **Acquisition and Sustainment Data Package (ASDP) Environmental Criteria Document (ECD) and Report**

Number: DI-ENVR-82540
AMSC Number: 10661
DTIC Applicable: No
Preparing Activity: AF-11 (AFLCMC/EN-EZ)
Applicable Forms: N/A

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Limitation: N/A
GIDEP Applicable: No
Project Number: ENVR-2026-005

Use/Relationship: The Environmental Criteria Document (ECD) shall provide a detailed characterization of the natural and induced environments to which the system, subsystem, zone, assembly, or part will be subjected throughout the life cycle. The principal uses are to (1) define the natural environments an air system will be exposed to for its service life, (2) contain the measured system environmental stress levels to be used for design criteria generation and qualification activities as required by MIL-STD-1530 Aircraft Structural Integrity Program (ASIP), MIL-STD-1798 Mechanical Equipment and Subsystems Integrity Program (MECSIP), MIL-HDBK-525 Guidance for the Design, Development, and Testing of Electrical Wiring Interconnection Systems (EWIS), MIL-STD-3024 Propulsion Systems Integrity Program, Avionics Integrity Program (AVIP), and MIL-STD-3063 Rotorcraft Structural Integrity Program (RSIP) (3) characterize the induced environments, including those generated by the material platform, and (4) identify the verification activities that will show compliance to the specification requirements and airworthiness certification criteria associated with air system environment in the natural and induced environments for its service life.

a. This Data Item Description (DID) is applicable to any contract pertaining to air vehicle development, modification, or when an Environmental Criteria Document is required.
b. This Data Item Description contains the format, content, and intended use information for the data deliverable resulting from the work task described in the solicitation.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions are as specified in the contract.

MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

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MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-961	Defense and Program-Unique Specifications Format and Content
MIL-STD-1530	Aircraft Structural Integrity Program (ASIP)
MIL-STD-1798	Mechanical Equipment and Subsystems Integrity Program
MIL-STD-3024	Propulsion System Integrity Program (PSIP)
MIL-STD-3063	Rotorcraft Structural Integrity Program (RSIP)
MIL-PRF-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance, NATO Code H-515
MIL-HDBK-310	Global Climatic Data for Developing Military Products
MIL-HDBK-525	Electrical Wiring Interconnect System (EWIS) Integrity Program
DI-SESS-82361	Acquisition and Sustainment Data Package (ASDP) Technical Report

(Copies of these documents can be obtained at https://assistca.dla.mil/.)

Army Regulation 70-38	Research, Development, Test and Evaluation of Materiel for Worldwide Use

(Copy of this regulation can be obtained at https://armypubs.army.mil/epubs.)

RTCA DO-160	Environmental Conditions and Test Procedures for Airborne Equipment

(Copy of this document can be obtained at https://www.rtca.org/.)

2. Format. The Environmental Criteria Document format shall be contractor selected. Each update report shall reflect the total environmental analysis and status.

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3. Content. The Environmental Criteria Document shall contain the following:

3.1 System description. This will consist of summary descriptions of the physical and functional characteristics of the air system and its components. Reference to more detailed system and component descriptions, including specifications and detailed review documentation shall be supplied when documentation is available. The capabilities, limitations, and interdependence of these components shall include identification of items and functions associated with safe operation of the air system. System block diagrams or functional flow diagrams shall be used to clarify system description. The intended operational usage of the system shall be identified including base and deployed locations, flight envelope, transportation, and storage. A Life Cycle Environmental Profile (LCEP), as defined by MIL-STD-810 Task 402, captures the necessary requirements to define intended operational usage for environment definition and characterization purposes.

3.2 Natural environments for the air system. The ECD shall present the intended natural environments the air system and its subsystems, zones, assemblies, and components will be exposed to for its service life. This definition is predicated on the intended concept of operations (CONOPS), operational mode summary (OMS), and mission profile (MP), including the information on base and deployed locations, flight envelope, transportation, and storage. Guidance for identifying environments is contained in MIL-HDBK-310, Global Climatic Data for Developing Military Products, MIL-STD-810, MIL-STD-461 Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems, Integrity Programs listed Use/Relationship section of the DID, Environmental Engineering Considerations and Laboratory Tests, and Army Regulation 70-38, Research, Development, Test and Evaluation of Materiel for Worldwide Use. For each natural environment, define the high and low values and anticipated environmental cycles appropriate to the CONOPS/OMS/MP.

The following shall be delivered: CONOPS/OMS/MP and the analysis that goes with environments below.

a. The environmental parameters are defined by:

    1) Extreme values

        a) For overall air system environment, if data other than that contained in MIL-HDBK-310 is used, provide substantiating information on validity of the data.

        b) Extreme values for certain subsystems are addressed by other standards (e.g., MIL-PRF-5606 for hydraulics).

    2) Transient values, rates, duration

        a) Addresses all operational modes (e.g. taxi, takeoff, climb, cruise, dogfighting, descent, landing).

    3) Times of exposure

    4) Exposure levels

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5) Spectrum indication

b. The natural environments that must be addressed for the intended CONOPS/OMS/MP are:

1) Temperature
2) Atmospheric pressure
3) Atmospheric density
4) Wind speed and wind shear
5) Humidity
6) Rainfall rate and water concentration
7) Blowing snow and snowload
8) Ice accretion
9) Hail size
10) Ozone concentration
11) Solar radiation
12) Sand and dust
13) Freeze-thaw cycles
14) Salt atmosphere and spray

3.3 Zone definitions. To simplify environmental parameter definitions for the air system and its subsystems, assemblies, and components, zones shall be defined for the air system. The ECD shall present the analytical and measured characterization of air system zones. When such information is based on estimates, it shall be replaced by actual values as they are determined. Examples of zones to be defined are below:

a. Temperature zones based on equipment cooling considerations as well as induced thermal environment from propulsion system and component operation

b. Pressurization of compartments

c. Hazard zones

d. Zones based on vibration spectra determination

e. Flammable fluid leakage zone

3.4 Integrity programs. Integrity Programs listed Use/Relationship section of the DID required data shall be traced and described with the information below. The ECD shall present analytical and measured characterization of the environmental stresses that the system, subsystem, zone, assembly, or part are expected to endure throughout their lifecycle. When such information is based on estimates, it shall be replaced by actual values as they are determined. Each update to

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the document shall reflect the total environment and status. The document shall be prepared based on the interpretation and evaluation of tests and analyses:

a. The document shall contain the following:
    1) Initial estimates of the considered environment of the system.
    2) Justification of how the variation of the environment is understood, measured, quantified, and accounted for.
    3) The number of measurements for which the average stress values and design achievements are determined as well as their characteristic variability expressed in terms of the standard deviation.
    4) Environmental parameters defined by (when applicable):
        a) Desired accuracy
        b) Extreme values
        c) Transient values, rates, duration
        d) Times of exposure
        e) Exposure levels
        f) Spectrum indication
        g) Support structures transmissivity values
    5) Standard publication procedures for test procedure descriptions. Procedures that differ from standard publications shall be thoroughly explained.
    6) For Contractor off the Shelf (COTS) or Government off The Shelf (GOTS):
        a) Discussion of parameters as part of a natural environmental interface or database.
        b) Presentation of qualification reports.
    7) Description of the flight/ground test programs and instrumentation for verifying the established environmental requirements.
    8) An overview of the environmental boundary.
    9) Environmental problem areas including indications of the need for greater environmental forecasting capability and areas requiring greater Research and Development (R&D) effort.
    10) Presentation of the status of environmental progress reports which shall contain the following:
        a) All changes to the established environmental requirements, testing procedures, or flight test program.
        b) A summarization of R&D environmental efforts and problem areas.
        c) A summarization of all environmental test data.

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11) Environmental parameters induced by equipment operation, transportation, routine testing, handling, and storage.

12) Description of the environment that the aircraft is exposed to regarding the type of exposure, rate of exposure, and exposure attributes.

b. Environmental considerations:

1) Ambient pressures, unpressurized compartments, cockpit pressures, and fuel tank pressures

2) Aircraft storage temperatures, internal compartment operating and non-operating temperatures, external surface temperatures, and electronic equipment cooling

3) Crew station ventilation, weapon bay ventilation, and engine bay ventilation

4) Acoustic vibration, noise vibration, and mechanical vibration, gun vibration

5) Acceleration loads, gyroscopic moments, and shock loads

6) Explosive atmosphere and fire protection zones

7) Humidity, moisture, and precipitation

8) Salt spray, gunfire gases, missile gases, sulfur and nitrogen oxides

9) Fungus, sand, and dust

10) Electromatic Interference (EMI), Electromatic Compatibility (EMC), lightning, and lasers

c. Documentation of new, analytically correlated test data in a revision to the ECD after any significant environmental test program is completed. These test programs shall include all of the requirements listed above, to include, aircraft acoustic and vibration flight testing, gun fire testing, stores acoustic and vibration flight testing, and engine ground run acoustic and vibration testing.

3.5 Induced environments for the air system. Induced environments are predominantly man-made or generated by the material platform. It also refers to any condition internal to material that results from the combination of natural environment forcing functions and the physical/chemical characteristics of the material itself. Definition of the induced environments is predicated on the intended CONOPS/OMS/MP, including the information on base and deployed locations, flight envelope, transportation, and storage for the air system and its subsystems, assemblies, and parts. The ECD shall present the analytical and measured characterization of the induced environments that the air system, subsystem, zone, assembly, or part are expected to endure throughout their service life. When such information is based on estimates, it shall be replaced by actual values as they are determined. Each update to the document shall reflect the total environment and status.

a. Air system considerations that affect induced environmental parameters include:

1) Intended altitude envelope

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2) Canopy thermal environment

3) Forced air cooling (forced air flow)

4) Ventilation in locations including crew station(s), weapon bay(s), engine bay(s), Auxiliary Power Unit (APU) bay, and equipment bay(s)

5) Platform acceleration, deceleration, and maneuver, including crash landing loads

6) Mechanical, acoustic, and gunfire vibration

7) Handling and transportation shocks

8) Fluid resistance

9) Water intrusion and collection, including open doors during operations/servicing and seepage through the air vehicle skin

b. The environmental parameters are defined by:

1) Extreme values

2) Transient values, rates, duration

a) Addresses all operational modes (e.g. taxi, takeoff, climb, cruise, dogfighting, descent, landing) for the air system, subsystem, zones, assemblies, and parts.

b) Also addresses transportation and shipping for subsystems, assemblies, and parts.

3) Times of exposure

4) Exposure levels

5) Spectrum indication

c. The ECD must tailor the application of induced environments for the air system and all subsystems, zones, assemblies and parts. Induced environments identified in MIL-STD-810 are listed below. Additional induced environments shall be identified depending on the air system and its usage:

1) Pressure

2) Temperature and temperature shock. This includes addressing equipment cooling considerations as well as induced thermal environment from propulsion system and component operation.

3) Solar radiation

4) Contamination by fluids

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5) Humidity, moisture, and precipitation. This includes effects of water intrusion and concentration.

6) Immersion

7) Fungus

8) Salt fog and corrosive environments. This includes gunfire gases, missile gases, sulfur and nitrogen oxides, and chemical, biological, radiological, and nuclear weapons/materials.

9) Explosive atmosphere and hazard zones

10) Acceleration. List all types of loads. The ECD shall address air vehicle, aerodynamic, maneuvering, crash loads, weapons load outs, etc. Loads must be determined via test versus using generic values. The ECD shall include load values for the air vehicle that would be used for qualification and verification for all components.

11) Vibration. Including mechanical, acoustic, and gunfire vibration. Vibration shall be determined via test versus using generic values. Where vehicles are divided into zones, vibration is distinct for each zone. The ECD includes vibration values for the air vehicle that would be used for qualification and verification for all components.

12) Shock. Including information on types of shock. Shock shall be determined via test versus using generic values. The ECD includes shock values for the air vehicle that would be used for qualification and verification for all components.

13) Acidic atmosphere

14) Vibro-acoustic/temperature for externally carried aircraft stores

3.6 Environmental requirement mapping. The natural and induced environment requirements are mapped out for the air system to the lowest level. Requirements can be allocated to zones, assemblies, components, or other level of the physical architecture. Safety criticality of the system, subsystem, assembly, or component is identified in the requirements mapping. This supports identification of the environmental assessment necessary to determine airworthiness of the air system. The mapping shall be correlated to the analysis and testing of the zones.

a. The air system includes the air vehicle plus the support systems for the air vehicle (e.g., communications, control, ground/surface/control station, launch and recovery, and support elements).

3.7 Verification activities. For each environmental requirement, the verification methodology is identified. This includes identification of necessary resources such as environmental testing or climatic laboratories.

a. Verification activities are denoted by the verification type as defined in MIL-STD-961, MIL-STD-461, MIL-STD-464, Integrity Programs listed Use/Relationship section

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of the DID, and MIL-HDBK-310. Verification types are examination, analysis, demonstration, and test (laboratory, ground, and flight).

b. Verification activities are included in the environmental requirement mapping noted in paragraph 3.6.

c. A requirements and verification mapping analysis with test plans, procedures, and reports shall be provided with Zones and system, subsystems, and components in the zones with reference to the requirement ID and verification artifact (test plans, procedures, and reports). This shall be in the form of a table with the components and test plan, procedures, and reports associated. There shall be an annotation if the system, subsystems, and components do not meet the requirement and reference the gap analysis in 3.8. The test plan, procedures, and verification reports shall also be delivered, and be in accordance with DI-SESS-82361 Acquisition and Sustainment Data Package (ASDP) Technical Report.

3.8 Gap analysis for requirements. Where the program deviates from published military standards for environmental requirements (e.g., use of DO-160 in lieu of MIL-STD-810 or use of an earlier revision of the standard), a gap analysis is included to address the differences. If a gap is found, the 3.7.c shall have a risk traced to the gap per MIL-STD-882 and the requirement(s).

End of DI-ENVR-82540.

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